# A Lieb–Robinson bound for the Kogut–Susskind Hamiltonian, with velocity $\propto Nc/g^2$

The Hamiltonian of the obligations map has exactly the structure for which a Lieb–Robinson bound survives an unbounded generator: its unbounded part is a sum of commuting single-link terms, and its interaction is a bounded finite-range sum of plaquette terms. Passing to the interaction picture generated by the electric term leaves a time-dependent bounded interaction of unchanged norm and range, so the standard theorem applies and gives, for observables $A$ on a set $X$ of links and $B$ on $Y$,

$$\big\|[A(t),B]\big\| \;\le\; 2\|A\|\,\|B\|\,|X|\,e^{-\left(d(X,Y)-v_{\mathrm{LR}}|t|\right)/a}, \qquad v_{\mathrm{LR}} \;\le\; C_{\mathrm{LR}}aJ/\hbar \;\le\; 16C_{\mathrm{LR}}Nc/g^2,$$

where $J = \max_\ell \sum_{p\ni\ell}\|w_p\| \le 16N\hbar c/(g^2a)$ on the periodic three-dimensional spatial cubic lattice. The finite constant $C_{\mathrm{LR}}$ depends on the graph metric and theorem norm, including four-link support factors; it is independent of volume and coupling. The earlier $C_{\mathrm{LR}} = 2e$ was not derived from the metadata-only source citations and is not certified here.

With a separately established gap and vacuum hypotheses the standard clustering theorem applies. Its bound retains the microscopic range; $\hbar v_{\mathrm{LR}}/\Delta$ alone is not a general correlation-length bound. A divergent velocity upper estimate proves no superluminality. Locality is also not by itself a gap-stability theorem.

## 1. The structure that makes the unbounded term harmless

Take the links as the sites of the lattice and write, as in the obligations map §1,

$$H = H_{\mathrm{E}} + W, \qquad H_{\mathrm{E}} = \frac{\hbar c\, g^2}{2a}\sum_{\ell\in\mathcal{E}}(-\Delta_\ell), \qquad W = \sum_{p\in\mathcal{P}} w_p, \quad w_p = \frac{2\hbar c}{a\, g^2}(N - \operatorname{Re}\operatorname{tr} U_p).$$

Three properties are all that the argument uses.

1. *The unbounded part is single-site and commuting.* $-\Delta_\ell$ acts on the single factor $L^2(G)$ of link $\ell$, and the terms for different links commute. Each generates a one-parameter unitary group $e^{-is c g^2 \Delta_\ell/(2a)}$ whose adjoint action preserves the algebra of that link.
2. *The interaction is bounded.* $0 \le w_p \le \frac{4N\hbar c}{ag^2}$ in operator norm for $SU(N)$, since $|\operatorname{Re}\operatorname{tr} U| \le N$; for a general compact $G$ in a faithful representation of dimension $\dim_{\mathrm{f}}$ replace $N$ by $\dim_{\mathrm{f}}$.
3. *The range is one plaquette.* Each $w_p$ acts on the four links of $p$; a given link lies in exactly four plaquettes in three spatial dimensions (two planes containing its direction, two plaquettes on either side in each). Hence

$$J := \max_\ell \sum_{p\ni\ell}\|w_p\| \;\le\; 4\cdot\frac{4N\hbar c}{ag^2} = \frac{16N\hbar c}{a\,g^2}.$$

## 2. The reduction and the bound

**Proposition 1.** Let $\tau_t(A) = e^{itH/\hbar}Ae^{-itH/\hbar}$. For $A$ supported on a set $X$ of links and $B$ on a disjoint set $Y$, with $d(X,Y)$ the lattice distance,

$$\big\|[\tau_t(A),B]\big\| \;\le\; 2\|A\|\,\|B\|\,|X|\,\exp\Big[\frac{v_{\mathrm{LR}}|t| - d(X,Y)}{a}\Big], \qquad v_{\mathrm{LR}} \le C_{\mathrm{LR}}aJ/\hbar.$$

*Proof.* Let $U_{\mathrm{E}}(t) = e^{-itH_{\mathrm{E}}/\hbar}$ and define the interaction-picture interaction $\tilde w_p(t) = U_{\mathrm{E}}(t)^* w_p U_{\mathrm{E}}(t)$. Since $H_{\mathrm{E}}$ is a sum of commuting single-link terms, $U_{\mathrm{E}}(t)$ factorizes over links and its adjoint action preserves each link algebra, so $\tilde w_p(t)$ is supported on the same four

links and $\|\tilde{w}_p(t)\| = \|w_p\|$ for all $t$. The full evolution factorizes as $e^{-itH/\hbar} = U_{\mathrm{E}}(t)\,\tilde{U}(t)$ with $\tilde{U}$ generated by the time-dependent bounded interaction $\tilde{W}(t) = \sum_p \tilde{w}_p(t)$, and conjugation by $U_{\mathrm{E}}(t)$ preserves supports and norms. The Lieb–Robinson theorem for time-dependent bounded finite-range interactions (Lieb–Robinson, Commun. Math. Phys. 28 (1972) 251; Nachtergaele–Sims, Commun. Math. Phys. 265 (2006) 119, in the form used for the clustering theorem; metadata level) applies to $\tilde{W}$ with the constant $J$ of Section 1, and transferring back through $U_{\mathrm{E}}$ gives the stated volume-uniform scaling. A numerical prefactor requires fixing the theorem norm and graph-distance convention, not only $J$. □

The same argument is the one used for oscillator lattices, where the unbounded on-site harmonic part is treated exactly and the bounded anharmonic interaction carries the velocity (Nachtergaele–Raz–Schlein–Sims, Commun. Math. Phys. 286 (2008) 1073; metadata level). The gauge constraint plays no role: the bound holds on the full Hilbert space and restricts to the physical subspace, which is invariant under $H$.

**Remark (what fails without the structure).** If the electric term coupled two links, its unboundedness would enter the iteration and no bound of this form would follow. The single-link form of $\sum_a E^a_\ell E^a_\ell$ is therefore load-bearing, and it is the same feature that made the strong-coupling hypothesis check of the T2 note work.

## 3. Clustering and the microscopic range

The strong-coupling theorem, with its infinite-volume construction and vacuum hypotheses, already gives exponential clustering. The standard gapped-Hamiltonian argument has a conservative range-dependent form

$$\xi \le C\left(a + \frac{\hbar v_{\mathrm{LR}}}{\Delta}\right).$$

The constant depends on the locality norm and metric. The range term cannot be discarded uniformly when the quotient tends to zero. With $\Delta \ge \gamma C_2 g^2 \hbar c/(2a)$, the algebraic consequence is

$$\frac{\hbar v_{\mathrm{LR}}}{\Delta} \le \frac{32 C_{\mathrm{LR}} a N}{\gamma C_2 g^4}.$$

For $SU(3)$ the same norm count gives $J \le 48\hbar c/(ag^2)$, $v_{\mathrm{LR}} \le 48 C_{\mathrm{LR}} c/g^2$ and $\hbar v_{\mathrm{LR}}/\Delta \le 72 C_{\mathrm{LR}} a/(\gamma g^4)$. The theorem-dependent $C_{\mathrm{LR}}$ is still unspecified.

This controls the dynamical quotient, not $\xi$ by itself. The earlier claim $\xi = O(ag^{-4})$ is withdrawn. A plaquette-tube weight of order $g^{-4d/a}$ instead has inverse decay length of order $\log g/a$. Nachtergaele–Sims, *CMP* **265** (2006) 119, and Hastings–Koma, *CMP* **265** (2006) 781, are retained at metadata level; their theorems have their own hypotheses and numerical constants.

## 4. Weak coupling and physical causality

The locality upper estimate becomes weak as $g \to 0$. Comparing that coefficient with $c$ determines no actual propagation speed. Continuum relativistic locality requires a reconstructed physical theory.

Even the formal box-crossing comparison distinguishes a bound from an attained speed. Substituting an order $Nc/g^2$ coefficient and the conditional zero-mode energy $g^{2/3}\hbar c/L$ compares $Lg^2/(Nc)$ with $L/(cg^{2/3})$. The first formal time is smaller when $g^{8/3} < N$, up to constants. This proves neither causal connection of the box on that time scale nor a nonzero-mode truncation theorem.

## 5. Gap stability is a separate input

Locality is one ingredient of several stability theorems. Their further ground-space and perturbation-norm conditions must be checked; a frustration-free/topological-order theorem cannot be applied merely because a locality bound is available. Finite-coupling Kogut–Susskind electric and magnetic terms have no common minimizing state. The strong-coupling stability argument uses its own product reference and bounded perturbation.

No general openness theorem for the whole gapped coupling set follows here. Naming frustration-freeness does not exhaust the missing conditions.

## 6. Consequence for STATE

Treating commuting unbounded single-link terms exactly leaves bounded plaquette interactions with unchanged supports and norms. This supplies volume-uniform locality of order $Nc/g^2$. Numerical prefactors, physical propagation, clustering lengths, stability and continuum limits retain separate hypotheses.